

MODULE 6

Q1 Explain Web Mining and its type and explain application

Web mining is the best type of practice for sifting through the vast amount of data in the system that is available on the World Wide Web to find and extract pertinent information as per requirements. One unique feature of web mining is its ability to deliver a wide range of required data types in the actual process. There are various elements of the web that lead to diverse methods for the actual mining process. For example, web pages are made up of text; they are connected by hyperlinks in the system or process; and web server logs allow for the monitoring of user behavior to simplify all the required systems. Combining all the required methods from data mining, machine learning, artificial intelligence, statistics, and information retrieval, web mining is an interdisciplinary field for the overall system. Analyzing user behavior and website traffic is the one basic type or example of web mining.

Applications of Web Mining

Web mining is the process of discovering patterns, structures, and relationships in web data. It involves using data mining techniques to analyze web data and extract valuable insights. The applications of web mining are wide-ranging and include:

- **Personalized marketing:** Web mining can be used to analyze customer behavior on websites and social media platforms. This information can be used to create personalized marketing campaigns that target customers based on their interests and preferences.
- **E-commerce:** Web mining can be used to analyze customer behavior on e-commerce websites. This information can be used to improve the user experience and increase sales by recommending products based on customer preferences.
- **Search engine optimization:** Web mining can be used to analyze search engine queries and search engine results pages (SERPs). This information can be used to improve the visibility of websites in search engine results and increase traffic to the website.
- **Fraud detection:** Web mining can be used to detect fraudulent activity on websites. This information can be used to prevent financial fraud, identity theft, and other types of online fraud.

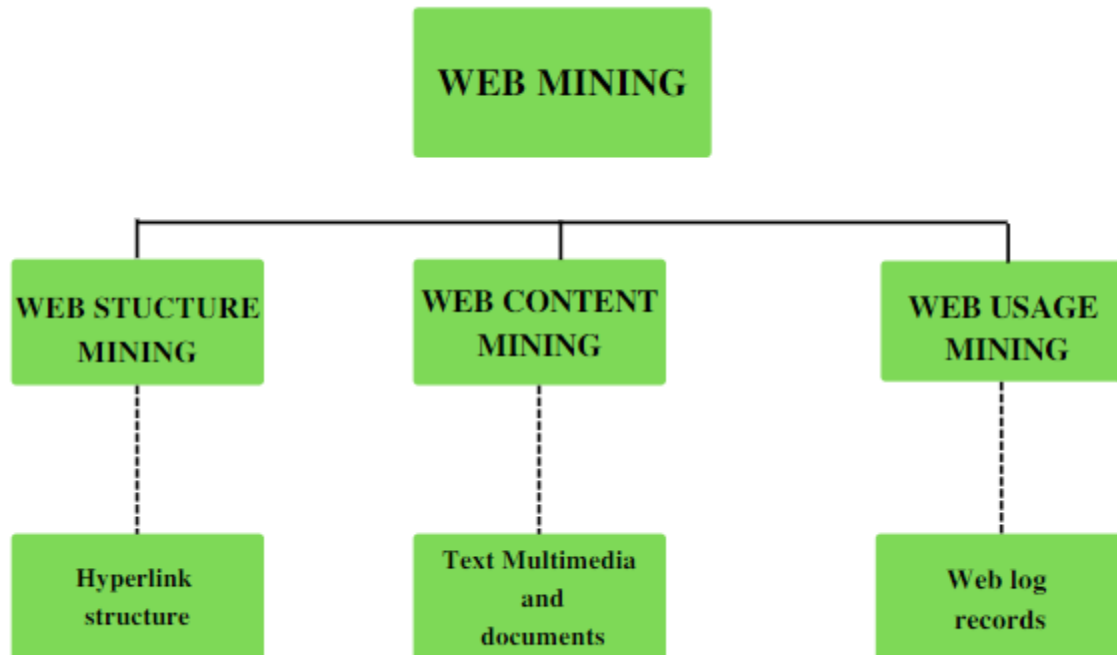
- **Sentiment analysis:** Web mining can be used to analyze social media data and extract sentiment from posts, comments, and reviews. This information can be used to understand customer sentiment towards products and services and make informed business decisions.
- **Web content analysis:** Web mining can be used to analyze web content and extract valuable information such as keywords, topics, and themes. This information can be used to improve the relevance of web content and optimize search engine rankings.
- **Customer service:** Web mining can be used to analyze customer service interactions on websites and social media platforms. This information can be used to improve the quality of customer service and identify areas for improvement.
- **Healthcare:** Web mining can be used to analyze health-related websites and extract valuable information about diseases, treatments, and medications. This information can be used to improve the quality of healthcare and inform medical research.

Process of Web Mining



Web Mining Process

Web mining can be broadly divided into three different types of techniques of mining: Web Content Mining, Web Structure Mining, and Web Usage Mining. These are explained as following below.



Categories of Web Mining

- **Web Content Mining:** Web content mining is the application of extracting useful information from the content of the web documents. Web content consist of several types of data – text, image, audio, video etc. Content data is the group of facts that a web page is designed. It can provide effective and interesting patterns about user needs. Text documents are related to text mining, machine learning and natural language processing. This mining is also known as text mining. This type of mining performs scanning and mining of the text, images and groups of web pages according to the content of the input.
- **Web Structure Mining:** Web structure mining is the application of discovering structure information from the web. The structure of the web graph consists of web pages as nodes, and hyperlinks as edges connecting related pages. Structure mining basically shows the structured summary of a particular website. It identifies relationship between web pages linked by information or direct link connection. To determine the connection between two commercial websites, Web structure mining can be very useful.
- **Web Usage Mining:** Web usage mining is the application of identifying or discovering interesting usage patterns from large data sets. And these patterns enable you to understand the user behaviors or something like that. In web usage mining, user access data on the web and collect data in form of logs. So, Web usage mining is also called log mining.

Q2.Explain Page Ranking algorithm

The PageRank algorithm is one of the most foundational and important concepts in **Web Mining**, specifically within **Web Structure Mining**. Developed by Google founders Larry Page and Sergey Brin, it revolutionized how search engines determine the importance and authority of a web page.

Here is a detailed explanation of the PageRank Algorithm and its technique.

PageRank Algorithm: The Concept

PageRank is an algorithm used to measure the **authority** or **importance** of a website or a page based on the structure of the web's hyperlinks.

The Core Philosophy (The "Voting" System)

PageRank works on the idea of **citation analysis** applied to the web:

A page is important if it is linked to by other important pages.

Imagine the World Wide Web as a massive democratic election:

- Every link from Page A to Page B is considered a **vote** of confidence for Page B.
- The **value of the vote** depends on the importance of the page casting the vote. A link from an authoritative site (like a major university or news site) counts for much more than a link from a relatively unknown blog.

The Random Surfer Model

PageRank models the behavior of a hypothetical "**random surfer**" who clicks on links endlessly:

1. The surfer starts on a random page.
2. At any page, the surfer chooses a link uniformly at random and clicks it.
3. The PageRank score of a page is the probability that the random surfer will eventually land on that specific page.

PageRank Technique and Calculation

The PageRank of a page A ($PR(A)$) is calculated iteratively, distributing the score of a linking page among all its outgoing links.

The Basic Formula (The Iterative Update)

The PageRank of a page A is the sum of the PageRanks of all pages (T_i) linking to A , divided by the number of outbound links ($L(T_i)$) on each linking page:

$$PR(A) = \sum_{T_i \in B_A} \frac{PR(T_i)}{L(T_i)}$$

- $PR(A)$: The PageRank score of page A .
- B_A : The set of pages that link to page A .
- $PR(T_i)$: The current PageRank score of a linking page (T_i).
- $L(T_i)$: The number of outbound links on page T_i .

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Query successful

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The PageRank of a page A is the sum of the PageRanks of all pages (Ti) linking to A, divided by the number of outbound links (L(Ti)) on each linking page:

$$PR(A) = \sum_{Ti \in BA} \frac{PR(Ti)}{L(Ti)}$$

- PR(A): The PageRank score of page A.
- BA: The set of pages that link to page A.
- PR(Ti): The current PageRank score of a linking page (Ti).
- L(Ti): The number of outbound links on page Ti.

The Damping Factor (Solving the "Dead End" Problem)

The basic formula above has two major flaws:

1. **Dangling Nodes (Dead Ends):** Pages with no outgoing links (dangling nodes) effectively absorb PageRank score, removing it from the system and causing other pages to lose rank over time.
2. **Rank Sinks:** A group of pages that only link to each other (a closed loop) trap PageRank within themselves, unfairly boosting their score.

To solve this, the **Damping Factor (d)** is introduced. The factor models the probability that the random surfer, instead of clicking a link, will suddenly get bored and **teleport** to a random page in the entire web. The typical value for **d** is **0.85**.

The Full PageRank Formula

$$PR(A) = (1 - d) + d \sum_{T_i \in B_A} \frac{PR(T_i)}{L(T_i)}$$

- **(1 - d) (The Teleportation Term):** This term (usually 0.15) ensures that every page has a minimum, baseline PageRank score. It represents the probability that the surfer teleports to a page *A* directly.
- **d ∑ . . . (The Link Term):** This term represents the probability (*d*) that the surfer follows a link and arrives at page *A*.